

Lyu Zizheng (Singapore Citizen)

Senior Product & Systems Engineer | Wise IC3 | Ex-ByteDance / TikTok

Email: lyuzizheng@gmail.com | Phone: +65 83423169 | Web: brabalawuka.cc | GitHub: github.com/Brabalawuka

SUMMARY

Product-minded Systems Engineer with 5+ years building mission-critical financial automation, high-concurrency IM infrastructure, and developer platforms. Passionate about engineering rigor, deterministic DAG workflows, and operational impact.

EXPERIENCE

WISE â Product Engineer 3 (Singapore) | Nov 2024 - Present

- Architected financial AI Automation & Evaluation Platform for last-mile payment linking without golden datasets.
- Implemented real-time live shadowing against PayOps, 99.5% accuracy gating, and 5% canary control failover.
- Automated 30,000+ complex payment defect cases/mo across heterogeneous partner rails, saving 80,000 GBP/mo.
- Re-engineered end-to-end global Compensation Platform with multi-party compliance reviews and double-approval controls.

MARIBANK (Sea Group) â Senior Backend Engineer (Singapore) | Apr 2024 - Jul 2024

- Engineered core digital banking credit engine (Cashloan & SME Termloan) enforcing distributed consistency under MAS rules.
- Built internal RAG-based customer support assistant leveraging LLM embeddings.

BONDEE â Senior Software Engineer (1st SWE in Singapore) | Sep 2023 - Apr 2024

- Established foundational SG cloud-native infrastructure, multi-tenant K8s, and Vector + Kafka logging pipelines.

BYTEDANCE / TIKTOK â Senior Backend Engineer (Singapore) | Jan 2021 - Sep 2023

- Overhauled TikTok IM Presence Engine: hybrid push-pull with Redis ZSET heartbeat gating, eliminating $O(N*M)$ write amplification.
- Redesigned conversation ranking & cold-start via active conversation hierarchy, achieving sub-100ms initialization.
- Sole engineer governing Overseas Location Platform (~100 Spark jobs, online Go services); awarded Spot Bonus & Top Rating (E).

EDUCATION & PROJECTS

Nanyang Technological University (NTU) â B.Eng. Honor in Computer Engineering (2017 - 2021)

- Full NTU Science and Engineering Undergraduate Merit Scholarship.
- U-Wave (Co-Founder): Scaled campus mobile app to 20,000 registered users and 4,000 DAU (Flutter + Spring Cloud).
- Side Projects: CanCan (reconciliation), CoffeeMode (cafe discovery), Our Village (decentralized community).